



```

name: SimulationBaltic
log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
log type: smcl
opened on: 25 Aug 2026, 10:09:50

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
> tion of Benefits for Older People in the Baltic States up to 2050 with data from SHA
> RE and Eurostat) fs(12) save(report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
2. clear all
3. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
> 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
4. use "sharew`num'_rel9-0-0_gv_imputations.dta"
5. keep if implicat==1
6. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
7. isid mergeid
8. drop _merge
9. merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
10. isid mergeid
11. drop _merge
12.
8 . if `num'==8{
13. gen Qyear=2020
14. }
15. if `num'==9{
16. gen Qyear=2022
17. }
18.
9 . keep if country==57 | country==48 | country==35
19.
10. save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
> \W`num'_final.dta", replace
20. }
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
> 0 ALL_datasets_stata
(214,780 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

      Result                                Number of obs
-----
Not matched                                0
Matched                                53,695   (_merge==3)

(label country already defined)
(label language already defined)

```

Result	Number of obs
Not matched	0
Matched	53,695

(`_merge==3`)

(47,632 observations deleted)

file **C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\W8_final.dta** saved

C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-

> 0 ALL datasets stata

(277,788 observations deleted)

(label **orienti** already defined)

(label **eurod** already defined)

(label **numeracy** already defined)

(label **gali** already defined)

(label **language** already defined)

(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	69,447

(`_merge==3`)

(label **country** already defined)

(label **language** already defined)

Result	Number of obs
Not matched	0
Matched	69,447

(`_merge==3`)

(61,831 observations deleted)

file **C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\W9_final.dta** saved

11.

12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label **country** already defined)

Result	Number of obs
Not matched	0
Matched	5,101

(`_merge==3`)

16. drop _merge

17. save "W8_final.dta", replace
file **W8_final.dta** saved

18.

19. clear all

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
21. use "W9_final.dta"
```

```
22. merge 1:1 mergeid using "weightslong.dta", keep(match)
    (label country already defined)
```

Result	Number of obs	
Not matched	0	
Matched	4,921	(_merge ==3)

```
23. drop _merge
```

```
24. save "W9_final.dta", replace
    file W9_final.dta saved
```

```
25.
```

```
26. clear all
```

```
27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
    C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
```

```
28. use "W8_final.dta"
```

```
29. append using "W9_final.dta"
    (variable psu was str10, now str11 to accommodate using data's values)
    (variable ssu was str6, now str13 to accommodate using data's values)
    (label pstat already defined)
    (label language already defined)
    (label country already defined)
    (label Hrespt already defined)
    (label pcskill already defined)
    (label IMPflag already defined)
    (label orienti already defined)
    (label bfi10 already defined)
    (label thospital already defined)
    (label satisfied already defined)
    (label often already defined)
    (label yedu already defined)
    (label fdistress already defined)
    (label writing already defined)
    (label reading already defined)
    (label hearing already defined)
    (label eyesight already defined)
    (label mstat already defined)
    (label sphus already defined)
    (label noyes already defined)
    (label hnrsc already defined)
    (label undersq already defined)
    (label clarif already defined)
    (label willans already defined)
    (label otrf already defined)
    (label empstat already defined)
    (label cjs already defined)
    (label isced already defined)
    (label gender already defined)
    (label num already defined)
    (label gali already defined)
    (label meat already defined)
    (label legeggs already defined)
    (label fruit already defined)
    (label numeracy already defined)
    (label numeracy2 already defined)
    (label eurod already defined)
    (label dairyp already defined)
    (label weekday already defined)
    (label specday already defined)
    (label dkrf already defined)
    (label dummi already defined)
```

```

(label zero_one already defined)
(label loneli already defined)
(label chronic2 already defined)
(label sphus2 already defined)
(label bmi2 already defined)
(label mobilit2 already defined)
(label mobilit3 already defined)
(label adl2 already defined)
(label iadl2 already defined)
(label phactiv already defined)

30.
31. save "dataset.dta", replace
    file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. label define isc 0 "None" 1 "Primary education" 2 "Lower secondary education" 3 "Upper
    > secondary education" 4 "Post-secondary non-tertiary education" 5 "Bachelor's or e
    > quivalent level" 6 "Master's or equivalent level/Doctoral or equivalent level"

40. label values isced isc

41. label variable isced "Education"

42.
43. generate income = thinc-ypen3

44.
45. egen id= group(mergeid)

46.
47. drop if cjs== -99
    (70 observations deleted)

48.
49. generate educ=0

50. replace educ=1 if isced==3 | isced==4
    (5,234 real changes made)

51. replace educ=2 if isced>=5
    (2,943 real changes made)

52. label define educL 0 "Less than primary, primary and lower secondary education (ISCE
    > D levels 0-2)" 1 "Upper secondary and post-secondary non-tertiary education (ISCED l
    > evels 3 and 4)" 2 "Tertiary education (ISCED levels 5-8)"

53. label values educ educL

54.
55. keep single ypen3 gender cjs age educ sphus country income gali id Qyear sphus

```

56.
 57. *>>>DESCRIPTIVES
 58.
 59. tabulate country

Country identifier	Freq.	Percent	Cum.
Estonia	5,526	55.53	55.53
Lithuania	2,423	24.35	79.87
Latvia	2,003	20.13	100.00
Total	9,952	100.00	

60. hist ypen3
 (bin=39, start=0, width=333.33333)

61.
 62. asdoc codebook single ypen3 gender cjs age educ sphus country income gali id Qyear,
 > compact save(report.doc) append

Variable	Obs	Unique	Mean	Min	Max	Label
single	9952	2	.4113746	0	1	Single
ypen3	9952	383	238.8071	0	13000	Disability pensions/benefits
gender	9952	2	1.631732	1	2	Gender
cjs	9952	6	1.94996	1	97	Current job situation
age	9952	64	69.68238	34	100	Age
educ	9952	3	1.117363	0	2	
sphus	9952	5	3.740856	1	5	Self-perceived health - US scale
country	9952	3	42.59295	35	57	Country identifier
income	9952	4456	10092.82	-.0000737	170698	
gali	9952	2	.588324	0	1	Limitation with activities
id	9952	5086	2559.018	1	5101	group (mergeid)
Qyear	9952	2	2020.98	2020	2022	

Click to Open File: [report.doc](#)

63.
 64. asdoc codebook, save(report.doc) append

country	Country identifier
Type: Numeric (byte) Label: country Range: [35,57] Unique values: 3 Units: 1 Missing .: 0/9,952	
Tabulation: Freq. Numeric Label 5,526 35 Estonia 2,423 48 Lithuania 2,003 57 Latvia	
single	Single
Type: Numeric (byte) Label: noyes Range: [0,1] Unique values: 2 Units: 1 Missing .: 0/9,952	

Tabulation:	Freq.	Numeric	Label
	5,858	0	No
	4,094	1	Yes

ypen3**Disability pensions/benefits**

Type: Numeric (**double**)

Range: [0,13000] Units: **1.000e-07**
Unique values: **383** Missing .: **0/9,952**

Mean: **238.807**
Std. dev.: **914.444**

Percentiles:	10%	25%	50%	75%	90%
	0	0	0	0	276

gender**Gender**

Type: Numeric (**byte**)
Label: **gender**

Range: [1,2] Units: **1**
Unique values: **2** Missing .: **0/9,952**

Tabulation:	Freq.	Numeric	Label
	3,665	1	Male
	6,287	2	Female

age**Age**

Type: Numeric (**int**)
Label: **num**, but **64** nonmissing values are not labeled

Range: [34,100] Units: **1**
Unique values: **64** Missing .: **0/9,952**

Examples: **60**
66
72
80

sphus**Self-perceived health - US scale**

Type: Numeric (**byte**)
Label: **sphus**

Range: [1,5] Units: **1**
Unique values: **5** Missing .: **0/9,952**

Tabulation:	Freq.	Numeric	Label
	154	1	Excellent
	416	2	Very good
	2,844	3	Good
	4,979	4	Fair
	1,559	5	Poor

gali**Limitation with activities**

Type: Numeric (**byte**)
Label: **gali**

Range: [0,1] Units: 1
 Unique values: 2 Missing .: 0/9,952

Tabulation:	Freq.	Numeric	Label
	4,097	0	Not limited
	5,855	1	Limited

cjs **Current job situation**

Type: Numeric (**byte**)
 Label: **cjs**

Range: [1,97] Units: 1
 Unique values: 6 Missing .: 0/9,952

Tabulation:	Freq.	Numeric	Label
	6,182	1	Retired
	2,948	2	Employed or self-employed
	241	3	Unemployed
	440	4	Permanently sick
	96	5	Homemaker
	45	97	Other

Qyear **(unlabeled)**

Type: Numeric (**float**)

Range: [2020,2022] Units: 1
 Unique values: 2 Missing .: 0/9,952

Tabulation:	Freq.	Value
	5,076	2020
	4,876	2022

income **(unlabeled)**

Type: Numeric (**float**)

Range: [-.00007374,170697.95] Units: 1.000e-20
 Unique values: 4,456 Missing .: 0/9,952

Mean: 10092.8
 Std. dev.: 8969.56

Percentiles:	10%	25%	50%	75%	90%
	1999.55	4920	7800	12895.6	20000

id **group (mergeid)**

Type: Numeric (**float**)

Range: [1,5101] Units: 1
 Unique values: 5,086 Missing .: 0/9,952

Mean: 2559.02
 Std. dev.: 1474.17

Percentiles:	10%	25%	50%	75%	90%
	513	1281.5	2563.5	3837.5	4596

educ **(unlabeled)**

Type: Numeric (**float**)
Label: **educL**

Range: [0,2]
Unique values: 3

Units: 1
Missing .: 0/9,952

Tabulation:	Freq.	Numeric	Label
	1,775	0	Less than primary, primary and lower secondary education (ISCED levels 0-2)
	5,234	1	Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
	2,943	2	Tertiary education (ISCED levels 5-8)

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- 65.
66. vioplot age, over(gender) over(country) scheme(s2color) ytitle("Number of respondent
> s", margin(small)) ylabel(30(10)100,angle(0) labszsize(small) grid)
67. graph export sample.png, replace height(2400)
(file **sample.png** not found)
file **sample.png** saved as PNG format
68. vioplot age if ypen3>0, over(gender) over(country) scheme(s2color) ytitle("Number of
> respondents", margin(small)) ylabel(30(10)100,angle(0) labszsize(small) grid)
69. graph export sampleDis.png, replace height(2400)
(file **sampleDis.png** not found)
file **sampleDis.png** saved as PNG format
- 70.
71. asdoc spearman single ypen3 gender cjs age educ sphus country income gali id Qyear s
> phus, save(report.doc) append

Number of observations = 9,952

>	income	single gali	ypen3 id	gender Qyear	cjs sphus	age	educ	sphus	country
	single	1.0000							
	ypen3	-0.0096	1.0000						
	gender	0.3013	-0.0183	1.0000					
	cjs	-0.1685	0.2522	-0.0665	1.0000				
	age	0.2603	-0.1646	0.0716	-0.7429	1.0000			
	educ	-0.1051	-0.0669	0.0455	0.1544	-0.2074	1.0000		
	sphus	0.0935	0.2086	-0.0050	-0.2445	0.3377	-0.2319	1.0000	
	country	-0.0445	-0.0944	-0.0115	0.0277	-0.0851	0.0267	-0.0232	1.0000
>	income	-0.4997	-0.1257	-0.1489	0.0389	-0.0763	0.2033	-0.1560	-0.3441
>	1.0000								
>	gali	0.0865	0.2264	0.0208	-0.2098	0.2909	-0.1698	0.5394	-0.0199
>	-0.1441	1.0000							
>	id	-0.0448	-0.0924	-0.0068	0.0230	-0.0729	0.0289	-0.0243	0.8979
>	-0.3002	-0.0243	1.0000						
>	Qyear	0.0286	-0.0337	0.0136	-0.0629	0.0851	0.0045	0.0147	0.0042
>	0.0789	0.0259	0.0038	1.0000					
>	sphus	0.0935	0.2086	-0.0050	-0.2445	0.3377	-0.2319	1.0000	-0.0232
>	-0.1560	0.5394	-0.0243	0.0147	1.0000				

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```

72.
73. *>>>ANALYSIS
74.
75. xtset id Qyear

Panel variable: id (unbalanced)
Time variable: Qyear, 2020 to 2022, but with gaps
Delta: 1 unit

76.
77. *-----
78. *ESTONIA
79.
80. *general model
81.
82. xttobit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
> selevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -13059.972
Iteration 1: Log likelihood = -8969.4753
Iteration 2: Log likelihood = -8848.5585
Iteration 3: Log likelihood = -8845.8105
Iteration 4: Log likelihood = -8845.8086
Iteration 5: Log likelihood = -8845.8086

```

Fitting full model:

```

Iteration 0: Log likelihood = -12748.166
Iteration 1: Log likelihood = -8956.6397
Iteration 2: Log likelihood = -8402.0038
Iteration 3: Log likelihood = -8347.2466
Iteration 4: Log likelihood = -8346.5547
Iteration 5: Log likelihood = -8346.5545

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -45249.222
Iteration 1: Log likelihood = -45212.213
Iteration 2: Log likelihood = -45211.856
Iteration 3: Log likelihood = -45211.856

```

Fitting full model:

```

Iteration 0: Log likelihood = -11974.249
Iteration 1: Log likelihood = -9486.0836 (not concave)
Iteration 2: Log likelihood = -8892.275
Iteration 3: Log likelihood = -8211.8878
Iteration 4: Log likelihood = -8201.1002
Iteration 5: Log likelihood = -8198.0475
Iteration 6: Log likelihood = -8198.0437
Iteration 7: Log likelihood = -8198.0437

```

Random-effects tobit regression

```

Limits: Lower = 0
Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: **mvaghermite**

Log likelihood = **-8198.0437**

```

Number of obs      = 5,526
Uncensored         = 767
Left-censored      = 4,759
Right-censored     = 0

```

```

Number of groups   = 2,837
Obs per group:
    min = 1
    avg = 1.9
    max = 2

```

Integration pts. = **12**

```

Wald chi2(6)       = 429.43
Prob > chi2        = 0.0000

```



```

>
>
>
>

```

```

> 2512.79
>      113.4388
>      22.15
>      0.000
>      2290.454
>      2735.126
>
> 1997.686
>      73.26852
>      27.27
>      0.000
>      1854.083
>      2141.29
>

```

```

> .6127313
>      .0271584
>
>      .5586039
>      .6647491
>

```

```

LR test of sigma_u=0: chibar2(01) = 297.02      Prob >= chibar2 = 0.000

```

83.

```

84. xttobit ypen3 i.gender age i.gali i.educ income if country==35, ll(0) iterate(25) ba
> selevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -13059.972
Iteration 1: Log likelihood = -8969.4753
Iteration 2: Log likelihood = -8848.5585
Iteration 3: Log likelihood = -8845.8105
Iteration 4: Log likelihood = -8845.8086
Iteration 5: Log likelihood = -8845.8086

```

Fitting full model:

```

Iteration 0: Log likelihood = -12748.166
Iteration 1: Log likelihood = -8956.6397
Iteration 2: Log likelihood = -8402.0038
Iteration 3: Log likelihood = -8347.2466
Iteration 4: Log likelihood = -8346.5547
Iteration 5: Log likelihood = -8346.5545

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -45249.222
Iteration 1: Log likelihood = -45212.213
Iteration 2: Log likelihood = -45211.856
Iteration 3: Log likelihood = -45211.856

```

Fitting full model:


```

Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> -653.254
>      217.7769
>      -3.00
>      0.003
>      -1080.089
>      -226.419
>      Tertiary education (ISCED levels 5-8) | -
> 1353.043
>      265.9793
>      -5.09
>      0.000
>      -1874.353
>      -831.7329
>
>      income | -
> .0763606
>      .0097108
>      -7.86
>      0.000
>      -.0953934
>      -.0573278
>
>      _cons |
> 5497.181
>      664.9515
>      8.27
>      0.000
>      4193.9
>      6800.462
>
>      /sigma_u |
> 2512.79
>      113.4388
>      22.15
>      0.000
>      2290.454
>      2735.126
>
>      /sigma_e |
> 1997.686
>      73.26852
>      27.27
>      0.000
>      1854.083
>      2141.29
>
>      rho |
> .6127313
>      .0271584
>      .5586039
>      .6647491
>
LR test of sigma_u=0: chibar2(01) = 297.02 Prob >= chibar2 = 0.000

```

```

85. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(CoreEE) replace
results.xls
dir : seeout

```

```

86. outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreEE) replace
   resultsNoStar.xls
   dir : seeout
87.
88. *for disability
89.
90. generate ageD=0

91. replace ageD=1 if age>=55 & age<=64
   (2,792 real changes made)

92. replace ageD=2 if age>=65 & age<=74
   (3,130 real changes made)

93. replace ageD=3 if age>=75 & age<=84
   (2,531 real changes made)

94. replace ageD=4 if age>=85
   (848 real changes made)

95. label define ageD1 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
   > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

96. label values ageD ageD1

97.
98. xtlogit ypen3 i.gender i.ageD i.gali i.educ income if country==35, ll(0) iterate(25)
   > baselevels

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -13059.972
Iteration 1: Log likelihood = -8969.4753
Iteration 2: Log likelihood = -8848.5585
Iteration 3: Log likelihood = -8845.8105
Iteration 4: Log likelihood = -8845.8086
Iteration 5: Log likelihood = -8845.8086

```

Fitting full model:

```

Iteration 0: Log likelihood = -12648.913
Iteration 1: Log likelihood = -8761.5091
Iteration 2: Log likelihood = -8249.2098
Iteration 3: Log likelihood = -8214.6742
Iteration 4: Log likelihood = -8214.5255
Iteration 5: Log likelihood = -8214.5255

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -45078.51
Iteration 1: Log likelihood = -45041.929
Iteration 2: Log likelihood = -45041.573
Iteration 3: Log likelihood = -45041.573

```

Fitting full model:

```

Iteration 0: Log likelihood = -11890.639
Iteration 1: Log likelihood = -9420.102 (not concave)
Iteration 2: Log likelihood = -8432.7335
Iteration 3: Log likelihood = -8207.1673
Iteration 4: Log likelihood = -8090.1718
Iteration 5: Log likelihood = -8088.846
Iteration 6: Log likelihood = -8088.8458

```

Random-effects tobit regression

```

Number of obs      = 5,526
Uncensored         = 767
Left-censored      = 4,759
Right-censored     = 0

```

```

Limits: Lower = 0
        Upper = +inf

```

Group variable: **id**
Random effects $u_i \sim \text{Gaussian}$

```
Number of groups = 2,837
Obs per group:
    min = 1
    avg = 1.9
    max = 2
```

Integration method: **mvaghermite**

Integration pts. = 12

Log likelihood = -8088.8458

```
Wald chi2(9)      = 601.85
Prob > chi2       = 0.0000
```

[illegible]

```

> 3124.31
>      201.2362
>      15.53
>      0.000
>      2729.894
>      3518.726
>
>      educ
>      Less than primary, primary and lower secondary education (ISCED levels 0-2)
>      0
>      (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) | -
> 581.0797
>      199.3518
>      -2.91
>      0.004
>      -971.8021
>      -190.3574
>      Tertiary education (ISCED levels 5-8) | -
> 1434.437
>      245.4567
>      -5.84
>      0.000
>      -1915.523
>      -953.3504
>
>      income | -
> .0629668
> .0087664
>      -7.18
>      0.000
>      -.0801486
>      -.045785
>
>      _cons | -
> 2380.608
>      364.0823
>      -6.54
>      0.000
>      -3094.197
>      -1667.02
>
>
>      /sigma_u |
> 2232.263
>      103.8189
>      21.50
>      0.000
>      2028.782
>      2435.744
>
>      /sigma_e |
> 1869.6
>      68.92764
>      27.12
>      0.000
>      1734.504
>      2004.696
>
>
>      rho |
> .5877284
> .0290497
>      .5300662
>      .6435539
>

```

LR test of sigma_u=0: chibar2(01) = 251.36 Prob >= chibar2 = 0.000


```

99. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityEE) append
results.xls
dir : seeout

```

```

100 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityEE) append
resultsNoStar.xls
dir : seeout

```

```

101
102 *for education
103
104 generate ageE=0

```

```

105 replace ageE=1 if age>=55 & age<=64
    (2,792 real changes made)

```

```

106 replace ageE=2 if age>=65
    (6,509 real changes made)

```

```

107 label define agel 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
> r than 64 years"

```

```

108 label values ageE agel

```

```

109
110 xttoibit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -13059.972
Iteration 1: Log likelihood = -8969.4753
Iteration 2: Log likelihood = -8848.5585
Iteration 3: Log likelihood = -8845.8105
Iteration 4: Log likelihood = -8845.8086
Iteration 5: Log likelihood = -8845.8086

```

Fitting full model:

```

Iteration 0: Log likelihood = -12655.689
Iteration 1: Log likelihood = -8767.8048
Iteration 2: Log likelihood = -8251.3306
Iteration 3: Log likelihood = -8216.2557
Iteration 4: Log likelihood = -8216.1028
Iteration 5: Log likelihood = -8216.1028

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -45081.832
Iteration 1: Log likelihood = -45047.601
Iteration 2: Log likelihood = -45047.303
Iteration 3: Log likelihood = -45047.303

```

Fitting full model:

```

Iteration 0: Log likelihood = -11888.4
Iteration 1: Log likelihood = -9407.8171 (not concave)
Iteration 2: Log likelihood = -8426.0436
Iteration 3: Log likelihood = -8190.8384
Iteration 4: Log likelihood = -8090.4197
Iteration 5: Log likelihood = -8089.7072
Iteration 6: Log likelihood = -8089.706
Iteration 7: Log likelihood = -8089.706

```

Random-effects tobit regression

```

Number of obs      = 5,526
Uncensored         = 767
Left-censored      = 4,759
Right-censored     = 0

```

```

Limits: Lower = 0
Upper = +inf

```

Group variable: **id**
 Random effects $u_i \sim \text{Gaussian}$

Number of groups = **2,837**
 Obs per group:
 min = **1**
 avg = **1.9**
 max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-8089.706**

Wald $\chi^2(7)$ = **597.41**
 Prob > χ^2 = **0.0000**

						ypen3	Co
	efficient	Std. err.	z	P> z	[95% con f. interval]		
						gender	
						Male	
>	0						
>	(base)						
>	62.35313	159.539	0.39	0.696	-250.3376	375.0438	
>							
>							
>							
>							
						ageE	
						Less than 55 years old	
>	0						
>	(base)						
						Between 55 and 64 years old	
>	592.9421	252.0415	2.35	0.019	98.94985	1086.934	
>							
>							
>							
>							
>							
						Older than 64 years	-
>	3159.591	262.9301	-12.02	0.000	-3674.924	-2644.257	
>							
>							
>							
>							
						gali	
						Not limited	
>	0						
>	(base)						
						Limited	
>	3114.176	201.3249	15.47	0.000	2719.587	3508.766	
>							
>							
>							
>							
						educ	
						Less than primary, primary and lower secondary education (ISCED levels 0-2)	
>	0						
>	(base)						
						Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)	-
>	544.0899	195.6608					
>							

>			-2.78	0.005	-927.5781	-160.6016		
>								
>								
>	1414.382						Tertiary education (ISCED levels 5-8)	-
>		244.2873						
>			-5.79					
>				0.000				
>					-1893.177			
>						-935.5882		
>							income	-
>	.0623023							
>		.0087611						
>			-7.11					
>				0.000				
>					-.0794737			
>						-.0451309		
>							_cons	-
>	2409.678							
>		363.5704						
>			-6.63					
>				0.000				
>					-3122.263			
>						-1697.093		
							/sigma_u	
>	2238.428							
>		103.8497						
>			21.55					
>				0.000				
>					2034.887			
>						2441.97		
>							/sigma_e	
>	1868.205							
>		68.9285						
>			27.10					
>				0.000				
>					1733.108			
>						2003.303		
							rho	
>	.5894256							
>		.0289619						
>					.5319203			
>						.6450675		

LR test of sigma_u=0: chibar2(01) = 252.79 Prob >= chibar2 = 0.000

```
111 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
    > title(EducationEE) append
    results.xls
    dir : seeout
```

```
112 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationEE) append
    resultsNoStar.xls
    dir : seeout
```

```

113
114 *for income
115
116 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==35, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -13059.972
Iteration 1: Log likelihood = -8969.4753
Iteration 2: Log likelihood = -8848.5585
Iteration 3: Log likelihood = -8845.8105
Iteration 4: Log likelihood = -8845.8086
Iteration 5: Log likelihood = -8845.8086

```

Fitting full model:

```

Iteration 0: Log likelihood = -12655.689
Iteration 1: Log likelihood = -8767.8048
Iteration 2: Log likelihood = -8251.3306
Iteration 3: Log likelihood = -8216.2557
Iteration 4: Log likelihood = -8216.1028
Iteration 5: Log likelihood = -8216.1028

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -45081.832
Iteration 1: Log likelihood = -45047.601
Iteration 2: Log likelihood = -45047.303
Iteration 3: Log likelihood = -45047.303

```

Fitting full model:

```

Iteration 0: Log likelihood = -11888.4
Iteration 1: Log likelihood = -9407.8171 (not concave)
Iteration 2: Log likelihood = -8426.0436
Iteration 3: Log likelihood = -8190.8384
Iteration 4: Log likelihood = -8090.4197
Iteration 5: Log likelihood = -8089.7072
Iteration 6: Log likelihood = -8089.706
Iteration 7: Log likelihood = -8089.706

```

Random-effects tobit regression

```

Limits: Lower = 0
Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

```

Number of obs      = 5,526
Uncensored         = 767
Left-censored      = 4,759
Right-censored     = 0

```

```

Number of groups   = 2,837
Obs per group:
    min = 1
    avg = 1.9
    max = 2

```

Integration method: mvaghermite

```

Integration pts.   = 12

```

Log likelihood = -8089.706

```

Wald chi2(7)      = 597.41
Prob > chi2       = 0.0000

```

```

> efficient
> Std. err.
>
> z
> P>|z|
> [95% con
> f. interval]

```

ypen3 | Co

gender |

```

> 0 Male |
> (base)
> 62.35313 Female |
> 159.539
> 0.39
> 0.696
> -250.3376
> 375.0438
>
> ageE
> Less than 55 years old |
> (base)
> 592.9421 Between 55 and 64 years old |
> 252.0415
> 2.35
> 0.019
> 98.94985
> 1086.934
> Older than 64 years | -
> 3159.591
> 262.9301
> -12.02
> 0.000
> -3674.924
> -2644.257
>
> gali
> Not limited |
> (base)
> 3114.176 Limited |
> 201.3249
> 15.47
> 0.000
> 2719.587
> 3508.766
>
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2) |
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) | -
> 544.0899
> 195.6608
> -2.78
> 0.005
> -927.5781
> -160.6016
> Tertiary education (ISCED levels 5-8) | -
> 1414.382
> 244.2873
> -5.79
> 0.000
> -1893.177
> -935.5882
>
> income | -
> .0623023
> .0087611
> -7.11
> 0.000
> -.0794737
> -.0451309
>
> _cons | -
> 2409.678
> 363.5704
> -6.63

```

```

>
>
>
0.000
-3122.263
-1697.093
-----
/|
> 2238.428
> 103.8497
> 21.55
> 0.000
> 2034.887
> 2441.97
> /sigma_e |
> 1868.205
> 68.9285
> 27.10
> 0.000
> 1733.108
> 2003.303
-----
|
> .5894256
> .0289619
> .5319203
> .6450675
-----
|
rho

```

LR test of sigma_u=0: chibar2(01) = 252.79 Prob >= chibar2 = **0.000**

```

117 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeEE) append
results.xls
dir : seeout

```

```

118 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeEE) append
resultsNoStar.xls
dir : seeout

```

```

119
120 *-----
121 *LITHUANIA
122
123 *general model
124
125 xttoibit ypen3 i.gender age i.gali i.educ income if country==48, ll(0) iterate(25) ba
> selevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4862.8488
Iteration 1: Log likelihood = -2762.1949
Iteration 2: Log likelihood = -2652.4901
Iteration 3: Log likelihood = -2648.5807
Iteration 4: Log likelihood = -2648.5779
Iteration 5: Log likelihood = -2648.5779

```

Fitting full model:

```

Iteration 0: Log likelihood = -4794.3696
Iteration 1: Log likelihood = -2581.4616
Iteration 2: Log likelihood = -2514.2583
Iteration 3: Log likelihood = -2511.8334
Iteration 4: Log likelihood = -2511.8237
Iteration 5: Log likelihood = -2511.8237

```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -19482.944
Iteration 1: Log likelihood = -19475.104
Iteration 2: Log likelihood = -19475.077
Iteration 3: Log likelihood = -19475.077
```

Fitting full model:

```
Iteration 0: Log likelihood = -4426.2983
Iteration 1: Log likelihood = -3129.6521 (not concave)
Iteration 2: Log likelihood = -2595.7117
Iteration 3: Log likelihood = -2441.8096
Iteration 4: Log likelihood = -2440.2936
Iteration 5: Log likelihood = -2439.7155
Iteration 6: Log likelihood = -2439.7155 (backed up)
Iteration 7: Log likelihood = -2439.7147
Iteration 8: Log likelihood = -2439.7147
```

Random-effects tobit regression

```
Limits: Lower = 0
        Upper = +inf
```

```
Group variable: id
Random effects u_i ~ Gaussian
```

```
Number of obs      = 2,423
Uncensored         = 218
Left-censored      = 2,205
Right-censored     = 0
```

```
Number of groups   = 1,235
Obs per group:
    min = 1
    avg = 2.0
    max = 2
```

Integration method: mvaghermite

```
Integration pts.   = 12
```

Log likelihood = -2439.7147

```
Wald chi2(6)       = 100.68
Prob > chi2        = 0.0000
```

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0							
> (base)							
						Female	-
> 1256.459	418.3478	-3.00	0.003	-2076.406	-436.5129		
						age	-
> 200.0298	27.40675	-7.30	0.000	-253.746	-146.3135		
						gali	
						Not limited	
> 0							
> (base)							
						Limited	
> 3058.35	422.143	7.24					

[illegible]

LR test of sigma_u=0: chibar2(01) = 144.22

```
Prob >= chibar2 = 0.000
```



```

126 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(CoreLT) append
  results.xls
  dir : seeout

127 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(CoreLT) append
  resultsNoStar.xls
  dir : seeout

128
129 *for disability
130
131 xttoibit ypen3 i.gender i.ageD i.gali i.educ income if country==48, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4862.8488
Iteration 1: Log likelihood = -2762.1949
Iteration 2: Log likelihood = -2652.4901
Iteration 3: Log likelihood = -2648.5807
Iteration 4: Log likelihood = -2648.5779
Iteration 5: Log likelihood = -2648.5779

```

Fitting full model:

```

Iteration 0: Log likelihood = -4773.3012
Iteration 1: Log likelihood = -2565.0925
Iteration 2: Log likelihood = -2484.8288
Iteration 3: Log likelihood = -2481.7728
Iteration 4: Log likelihood = -2481.7631
Iteration 5: Log likelihood = -2481.7631

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -19463.113
Iteration 1: Log likelihood = -19454.667
Iteration 2: Log likelihood = -19454.636

```

Fitting full model:

```

Iteration 0: Log likelihood = -4420.6504
Iteration 1: Log likelihood = -2956.5443 (not concave)
Iteration 2: Log likelihood = -2466.5563 (not concave)
Iteration 3: Log likelihood = -2451.0076
Iteration 4: Log likelihood = -2421.5766
Iteration 5: Log likelihood = -2420.0106
Iteration 6: Log likelihood = -2419.9666
Iteration 7: Log likelihood = -2419.9666 (backed up)
Iteration 8: Log likelihood = -2419.9661
Iteration 9: Log likelihood = -2419.9661

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: mvaghermite

Log likelihood = -2419.9661

```

Number of obs      = 2,423
Uncensored         = 218
Left-censored      = 2,205
Right-censored     = 0

```

```

Number of groups   = 1,235
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration pts. = 12

```

Wald chi2(9)       = 122.93
Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

```

```

> 0
> (base)
> 1226.406
> 405.9354
> -3.02
> 0.003
> -2022.025
> -430.7874
>
> 0
> (base)
> 1067.646
> 771.8018
> 1.38
> 0.167
> -445.058
> 2580.35
> 2545.243
> 873.6784
> -2.91
> 0.004
> -4257.622
> -832.8652
> 3972.701
> 981.3983
> -4.05
> 0.000
> -5896.206
> -2049.195
> 3319.867
> 1136.426
> -2.92
> 0.003
> -5547.222
> -1092.513
>
> 0
> (base)
> 3023.438
> 412.2912
> 7.33
> 0.000
> 2215.362
> 3831.513
>
> 0
> (base)

```

```

> ypen3 | Co
> gender
> Male |
> Female | -
> aged
> Less than 55 years old |
> Between 55 and 64 years old |
> Between 65 and 74 years old | -
> Between 75 and 84 years old | -
> Older than 85 years | -
> gali
> Not limited |
> Limited |
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)

```

```

Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4) |
> 396.2017
> 698.4553
> 0.57
> 0.571
> -972.7456
> 1765.149
> Tertiary education (ISCED levels 5-8) | -
> 140.2608
> 727.7094
> -0.19
> 0.847
> -1566.545
> 1286.023
> income | -
> .1087979
> .0271342
> -4.01
> 0.000
> -.16198
> -.0556158
> _cons | -
> 5421.922
> 1095.619
> -4.95
> 0.000
> -7569.295
> -3274.549
|-----|
| /sigma_u |
> 3647.767
> 297.3149
> 12.27
> 0.000
> 3065.04
> 4230.493
> /sigma_e |
> 2215.556
> 166.4085
> 13.31
> 0.000
> 1889.401
> 2541.71
|-----|
| rho |
> .7305124
> .0413152
> .6440207
> .8049676
|-----|

```

```

LR test of sigma_u=0: chibar2(01) = 123.59 Prob >= chibar2 = 0.000

```

```

132 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityLT) append
results.xls
dir : seeout

```

```

133 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLT) append
    resultsNoStar.xls
    dir : seeout

```

```
134
```

```
135 *for education
```

```
136
```

```

137 xtlogit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
    > baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4862.8488
Iteration 1: Log likelihood = -2762.1949
Iteration 2: Log likelihood = -2652.4901
Iteration 3: Log likelihood = -2648.5807
Iteration 4: Log likelihood = -2648.5779
Iteration 5: Log likelihood = -2648.5779

```

Fitting full model:

```

Iteration 0: Log likelihood = -4775.9907
Iteration 1: Log likelihood = -2566.0898
Iteration 2: Log likelihood = -2486.4369
Iteration 3: Log likelihood = -2483.2935
Iteration 4: Log likelihood = -2483.2848
Iteration 5: Log likelihood = -2483.2848

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -19465.758
Iteration 1: Log likelihood = -19457.753
Iteration 2: Log likelihood = -19457.725

```

Fitting full model:

```

Iteration 0: Log likelihood = -4422.6177
Iteration 1: Log likelihood = -2966.526 (not concave)
Iteration 2: Log likelihood = -2467.0794
Iteration 3: Log likelihood = -2428.6416 (backed up)
Iteration 4: Log likelihood = -2426.2572
Iteration 5: Log likelihood = -2422.3491
Iteration 6: Log likelihood = -2422.3393
Iteration 7: Log likelihood = -2422.3387
Iteration 8: Log likelihood = -2422.3383
Iteration 9: Log likelihood = -2422.3383

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

Integration method: mvaghermite

Log likelihood = -2422.3383

```

Number of obs      = 2,423
Uncensored         = 218
Left-censored      = 2,205
Right-censored     = 0

```

```

Number of groups   = 1,235
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration pts. = 12

```

Wald chi2(7)       = 121.50
Prob > chi2        = 0.0000

```

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

> ypen3 | Co
> 0
> (base)
> 1259.502
> 402.4633
> -3.13
> 0.002
> -2048.316
> -470.6886
>
> ageE
> Less than 55 years old
> 0
> (base)
> 1063.061
> 776.7657
> 1.37
> 0.171
> -459.3717
> 2585.494
> Older than 64 years | -
> 3051.484
> 852.7176
> -3.58
> 0.000
> -4722.78
> -1380.188
>
> gali
> Not limited
> 0
> (base)
> 3033.1
> 413.8208
> 7.33
> 0.000
> 2222.026
> 3844.174
>
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 805.6156
> 639.0382
> 1.26
> 0.207
> -446.8762
> 2058.107
> Tertiary education (ISCED levels 5-8)
> 232.6883
> 675.2926
> 0.34
> 0.730
> -1090.861
> 1556.237

```

```

> .1103631                                income | -
>      .0273208
>      -4.04
>      0.000
>      -.1639108
>      -.0568153
>      _cons | -
> 5793.143
>      1082.132
>      -5.35
>      0.000
>      -7914.083
>      -3672.203
+-----+
>      /sigma_u |
> 3621.426
>      292.6449
>      12.37
>      0.000
>      3047.852
>      4194.999
>      /sigma_e |
> 2250.038
>      167.5251
>      13.43
>      0.000
>      1921.695
>      2578.381
+-----+
>      rho |
> .7214854
>      .0414316
>      .6351044
>      .7964807
+-----+

```

LR test of sigma_u=0: chibar2(01) = 121.89 Prob >= chibar2 = 0.000

```

138 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(EducationLT) append
  results.xls
  dir : seeout

```

```

139 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLT) append
  resultsNoStar.xls
  dir : seeout

```

```

140
141 *for income
142
143 xttobit ypen3 i.gender i.ageE i.gali i.educ income if country==48, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -4862.8488
Iteration 1: Log likelihood = -2762.1949
Iteration 2: Log likelihood = -2652.4901
Iteration 3: Log likelihood = -2648.5807
Iteration 4: Log likelihood = -2648.5779
Iteration 5: Log likelihood = -2648.5779

```

Fitting full model:

```
> efficient                                     ypen3 | Co
>      Std. err.                               z       P>|z|
>                                         [95% con
>                                           f. interval]
+-----+-----+-----+-----+-----+-----+-----+
>                                gender          Male | 
>    0      (base)                                  -
> 1259.502                                Female   | 
>     402.4633                                   -
>           -3.13                                 -
>                0.002                             -
>                     -2048.316                      -
>                         -470.6886                    -
>                                                     ageE
>                   Less than 55 years old         | 
>    0      (base)                                    -
> 1063.061                                Between 55 and 64 years old | 
>     776.7657
```



```

Number of obs      =    2,003
   Uncensored      =     132
   Left-censored   =    1,871
   Right-censored   =         0

```

Group variable: **id**
 Random effects $u_i \sim \text{Gaussian}$

Number of groups = **1,014**
 Obs per group:
 min = **1**
 avg = **2.0**
 max = **2**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-1431.668**

Wald $\chi^2(6)$ = **94.04**
 Prob > χ^2 = **0.0000**

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0							
> (base)							
> 641.0127	402.2035	-1.59	0.111	-1429.317	147.2917	Female	-
						age	
> 217.7174	29.76099	-7.32	0.000	-276.0479	-159.3869		-
						gali	
						Not limited	
> 0							
> (base)							
> 4136.484	584.2676	7.08	0.000	2991.34	5281.627	Limited	
						educ	
						Less than primary, primary and lower secondary education (ISCED levels 0-2)	
> 0							
> (base)							
						Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)	
> 596.2011	616.5269	0.97	0.334	-612.1695	1804.572		
						Tertiary education (ISCED levels 5-8)	-
> 638.7354	817.3516	-0.78	0.435	-2240.715	963.2443		

```

> -.219487                                     income |
>      .0454556
>      -4.83
>      0.000
>      -.3085783
>      -.1303956
>
> 6995.528                                     _cons |
>      1787.025
>      3.91
>      0.000
>      3493.023
>      10498.03
>
+-----+-----+
> 2633.705                                     /sigma_u |
>      288.8006
>      9.12
>      0.000
>      2067.666
>      3199.744
>
> 2104.673                                     /sigma_e |
>      191.4314
>      10.99
>      0.000
>      1729.475
>      2479.872
>
+-----+-----+
> .6102737                                     rho |
>      .0643687
>      .4805192
>      .7287097
>
+-----+-----+

```

LR test of sigma_u=0: chibar2(01) = 50.13 Prob >= chibar2 = 0.000

```

153 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(Core) append
  results.xls
  dir : seeout

154 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(Core) append
  resultsNoStar.xls
  dir : seeout

155
156 *for disability
157
158 xttobit ypen3 i.gender i.ageD i.gali i.educ income if country==57, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```
Iteration 0: Log likelihood = -3506.908
Iteration 1: Log likelihood = -1552.0409
Iteration 2: Log likelihood = -1457.4938
Iteration 3: Log likelihood = -1445.8596
Iteration 4: Log likelihood = -1445.3313
Iteration 5: Log likelihood = -1445.2677
Iteration 6: Log likelihood = -1445.2581
Iteration 7: Log likelihood = -1445.2563
Iteration 8: Log likelihood = -1445.2559
Iteration 9: Log likelihood = -1445.2559
Iteration 10: Log likelihood = -1445.2558
```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -15445.47
Iteration 1: Log likelihood = -15441.736
Iteration 2: Log likelihood = -15441.727
```

Fitting full model:

```
Iteration 0: Log likelihood = -3290.4204
Iteration 1: Log likelihood = -1565.0715 (not concave)
Iteration 2: Log likelihood = -1487.9781
Iteration 3: Log likelihood = -1434.153
Iteration 4: Log likelihood = -1421.347
Iteration 5: Log likelihood = -1420.921
Iteration 6: Log likelihood = -1420.885
Iteration 7: Log likelihood = -1420.88
Iteration 8: Log likelihood = -1420.879
Iteration 9: Log likelihood = -1420.8788
Iteration 10: Log likelihood = -1420.8787
Iteration 11: Log likelihood = -1420.8787
```

Random-effects tobit regression

```
Limits: Lower = 0
        Upper = +inf
```

```
Group variable: id
Random effects u_i ~ Gaussian
```

```
Number of obs      = 2,003
Uncensored         = 132
Left-censored      = 1,871
Right-censored     = 0
```

```
Number of groups   = 1,014
Obs per group:
    min = 1
    avg = 2.0
    max = 2
```

Integration method: mvaghermite

```
Integration pts.   = 12
```

Log likelihood = -1420.8787

```
Wald chi2(9)       = 94.08
Prob > chi2        = 0.0000
```

						ypen3	Co
> efficient							
> Std. err.							
> z							
> P> z							
> [95% con							
> f. interval]							
						gender	
						Male	
> 0							
> (base)							
						Female	-
> 341.5978							
> 399.6168							
> -0.85							
> 0.393							
> -1124.832							
> 441.6368							


```

> 5421.555
>          990.1776
>          -5.48
>          0.000
>          -7362.267
>          -3480.842
+-----+
                                     /sigma_u |
> 2570.351
>          279.8937
>          9.18
>          0.000
>          2021.769
>          3118.933
+-----+
                                     /sigma_e |
> 2041.861
>          189.3496
>          10.78
>          0.000
>          1670.743
>          2412.98
+-----+
                                     rho |
>    .6131
>          .0653765
>          .4811277
>          .7330748
+-----+
LR test of sigma_u=0: chibar2(01) = 48.75          Prob >= chibar2 = 0.000

```

```

159 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(DisabilityLV) append
results.xls
dir : seeout

```

```

160 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(DisabilityLV) append
resultsNoStar.xls
dir : seeout

```

```

161
162 *for education
163
164 xttoibit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
> baselevels tobit

```

Fitting comparison model:

Fitting constant-only model:

```

Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126

```

Fitting full model:

```

Iteration 0: Log likelihood = -3511.125
Iteration 1: Log likelihood = -1555.4342
Iteration 2: Log likelihood = -1464.3483
Iteration 3: Log likelihood = -1455.0028
Iteration 4: Log likelihood = -1454.8874
Iteration 5: Log likelihood = -1454.8873

```

Obtaining starting values for full model:

```

Iteration 0: Log likelihood = -15450.207
Iteration 1: Log likelihood = -15446.684
Iteration 2: Log likelihood = -15446.676
Iteration 3: Log likelihood = -15446.676

```

Fitting full model:

```

Iteration 0: Log likelihood = -3293.0056
Iteration 1: Log likelihood = -1558.0001 (not concave)
Iteration 2: Log likelihood = -1483.8455
Iteration 3: Log likelihood = -1440.8292
Iteration 4: Log likelihood = -1431.4216
Iteration 5: Log likelihood = -1431.0339
Iteration 6: Log likelihood = -1431.0316
Iteration 7: Log likelihood = -1431.0316

```

Random-effects tobit regression

```

Limits: Lower = 0
        Upper = +inf

```

```

Group variable: id
Random effects u_i ~ Gaussian

```

```

Number of obs      = 2,003
Uncensored         = 132
Left-censored      = 1,871
Right-censored     = 0

```

```

Number of groups   = 1,014
Obs per group:
    min = 1
    avg = 2.0
    max = 2

```

Integration method: mvaghermite

Integration pts. = 12

Log likelihood = -1431.0316

```

Wald chi2(7)      = 98.95
Prob > chi2       = 0.0000

```

						ypen3	Co
> efficient	Std. err.	z	P> z	[95% con	f. interval]		
						gender	
						Male	
> 0							
> (base)							
						Female	-
> 433.5706	394.5748	-1.10	0.272	-1206.923	339.7818		
						ageE	
						Less than 55 years old	
> 0							
> (base)							
						Between 55 and 64 years old	
> 383.5329	601.6318	0.64	0.524	-795.6438	1562.71		
						Older than 64 years	
> -3431.92	739.1043	-4.64	0.000	-4880.537			

```

>
>                                     -1983.302
>                                     gali
>                                     Not limited
>
>      0
>      (base)
>
>                                     Limited
> 3710.976
>      551.8659
>      6.72
>      0.000
>      2629.338
>      4792.613
>
>                                     educ
>      Less than primary, primary and lower secondary education (ISCED levels 0-2)
>      0
>      (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 688.1303
>      582.771
>      1.18
>      0.238
>      -454.0798
>      1830.34
>      Tertiary education (ISCED levels 5-8)
> 468.8768
>      775.7078
>      -0.60
>      0.546
>      -1989.236
>      1051.483
>
>                                     income
> .1964301
>      .0446491
>      -4.40
>      0.000
>      -.2839408
>      -.1089194
>
>                                     _cons
> 5825.039
>      996.1594
>      -5.85
>      0.000
>      -7777.476
>      -3872.603
>
>                                     /sigma_u
> 2540.898
>      275.857
>      9.21
>      0.000
>      2000.228
>      3081.568
>
>                                     /sigma_e
> 2131.659
>      194.8313
>      10.94
>      0.000
>      1749.796
>      2513.521
>
>                                     rho
> .5869169
>      .064998
>      .4571951
>      .707723

```


LR test of sigma_u=0: chibar2(01) = 47.71

Prob >= chibar2 = 0.000

```
165 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(EducationLV) append
  results.xls
  dir : seeout
```

```
166 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(EducationLV) append
  resultsNoStar.xls
  dir : seeout
```

167

168 *for income

169

```
170 xttoibit ypen3 i.gender i.ageE i.gali i.educ income if country==57, ll(0) iterate(25)
> baselevels tobit
```

Fitting comparison model:

Fitting constant-only model:

```
Iteration 0: Log likelihood = -3579.9263
Iteration 1: Log likelihood = -1698.9624
Iteration 2: Log likelihood = -1616.2797
Iteration 3: Log likelihood = -1616.1126
Iteration 4: Log likelihood = -1616.1126
```

Fitting full model:

```
Iteration 0: Log likelihood = -3511.125
Iteration 1: Log likelihood = -1555.4342
Iteration 2: Log likelihood = -1464.3483
Iteration 3: Log likelihood = -1455.0028
Iteration 4: Log likelihood = -1454.8874
Iteration 5: Log likelihood = -1454.8873
```

Obtaining starting values for full model:

```
Iteration 0: Log likelihood = -15450.207
Iteration 1: Log likelihood = -15446.684
Iteration 2: Log likelihood = -15446.676
Iteration 3: Log likelihood = -15446.676
```

Fitting full model:

```
Iteration 0: Log likelihood = -3293.0056
Iteration 1: Log likelihood = -1558.0001 (not concave)
Iteration 2: Log likelihood = -1483.8455
Iteration 3: Log likelihood = -1440.8292
Iteration 4: Log likelihood = -1431.4216
Iteration 5: Log likelihood = -1431.0339
Iteration 6: Log likelihood = -1431.0316
Iteration 7: Log likelihood = -1431.0316
```

Random-effects tobit regression

Limits: Lower = 0
Upper = +inf

Group variable: id
Random effects u_i ~ Gaussian

Integration method: mvaghermite

Log likelihood = -1431.0316

Number of obs = 2,003
Uncensored = 132
Left-censored = 1,871
Right-censored = 0

Number of groups = 1,014
Obs per group:
min = 1
avg = 2.0
max = 2

Integration pts. = 12

Wald chi2(7) = 98.95
Prob > chi2 = 0.0000

```

> efficient
> Std. err.
> z
> P>|z|
> [95% con
> f. interval]

> ypen3 | Co
>
> 0
> (base)
> 433.5706
> 394.5748
> -1.10
> 0.272
> -1206.923
> 339.7818
>
> ageE
> Less than 55 years old
> 0
> (base)
> 383.5329
> 601.6318
> 0.64
> 0.524
> -795.6438
> 1562.71
> Older than 64 years
> -3431.92
> 739.1043
> -4.64
> 0.000
> -4880.537
> -1983.302
>
> gali
> Not limited
> 0
> (base)
> 3710.976
> 551.8659
> 6.72
> 0.000
> 2629.338
> 4792.613
>
> educ
> Less than primary, primary and lower secondary education (ISCED levels 0-2)
> 0
> (base)
> Upper secondary and post-secondary non-tertiary education (ISCED levels 3 and 4)
> 688.1303
> 582.771
> 1.18
> 0.238
> -454.0798
> 1830.34
> Tertiary education (ISCED levels 5-8)
> 468.8768
> 775.7078
> -0.60
> 0.546
> -1989.236
> 1051.483

```

```

> .1964301                                income | -
>      .0446491
>      -4.40
>      0.000
>      -.2839408
>      -.1089194
>
> 5825.039                                _cons | -
>      996.1594
>      -5.85
>      0.000
>      -7777.476
>      -3872.603
>
+-----+
> 2540.898                                /sigma_u |
>      275.857
>      9.21
>      0.000
>      2000.228
>      3081.568
>
> 2131.659                                /sigma_e |
>      194.8313
>      10.94
>      0.000
>      1749.796
>      2513.521
>
+-----+
> .5869169                                rho |
>      .064998
>      .4571951
>      .707723
>
+-----+

```

LR test of sigma_u=0: chibar2(01) = 47.71 Prob >= chibar2 = 0.000

```

171 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(IncomeLV) append
results.xls
dir : seeout

```

```

172 outreg2 using resultsNoStar.xls, excel dec(3) ctitle(IncomeLV) append
resultsNoStar.xls
dir : seeout

```

```

173
174 log close SimulationBaltic
      name: SimulationBaltic
      log: C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
log type: smcl
closed on: 25 Aug 2026, 10:11:02

```